Weight Manager · Development Document (English)

Weight Manager · v1.8.0 · 2026-08-14 · Compatible with Blender 4.2+ / 5.x

The full journey from v1.0 to v1.8.0, comparison with C4D / Maya, technical architecture, and future directions.
Chinese version: 开发文档.md. Project root: C:\Users\hasee\blender-weight-manager

1. Overview

Weight Manager is a Blender addon that turns weight painting from "Weight Paint mode + brush strokes" into "sidebar panel + sliders + numbers", closely modeled on Cinema 4D's Weight Manager panel while absorbing the Maya Component Editor's ability to "read and edit per-point numbers precisely".

- Form: a "Weight Mgr" tab in the 3D viewport's right sidebar (N panel)
- Compatibility: Blender 4.2+ / 5.x, every API difference is handled
- Current version: v1.8.0 (2026-08-14)
- Deliverables: classic addon zip + official extension format (blender_manifest.toml, publishable to extensions.blender.org)

Why it exists: Blender's native weight tools are "brush + color ramp" — you scrub back and forth with no numeric feedback. C4D's Weight Manager achieves precise control with "pick a joint in a panel + set a value with a slider". This addon brings that interaction into Blender, then adds Maya-style per-point numeric tables.

2. Development history (10 iterations)

Development from v1.0 to v1.8.0 went through 10 rounds, each driven by "user feedback → find the gap → rework → headless verification". No GUI testing was possible — everything relied on background (headless) test scripts plus code self-review.

v1.0 · Baseline (Round 1)

Core panel features modeled on C4D:

- Vertex Groups (bones) list: a real list widget + lock icon ([lock]), click to switch the current group
- Auto Weight: a slider that paints the selected vertices in real time
- Commands: Invert / Mirror
- Select by weight (Fill-Selection style)
- Integration with the Fill Select addon: one-click fill selection

Round 1 also tried a vertical modal "big button" painter (WeightOT_DragPaint); the UX was bad and it was scrapped entirely.

v1.1 · Auto Weight four modes (Round 2)

Feedback: "doesn't feel like C4D — C4D has a bar you can drag", plus the requirement that direction must not matter, only distance:

- auto_weight_mode enum, four options: ABSOLUTE / ADD / SUBTRACT / SMOOTH (ABSOLUTE default)
- ABSOLUTE: native wide horizontal slider, jump-to-value (pure set mode)
- ADD/SUBTRACT/SMOOTH: custom modal drag operator WeightOT_DragAccumulate, using per-tick increments (dx = mouse_region_x - last_x), dist = |dx|/100 * Strength — direction-independent, distance-only accumulation
- Also an "Apply once (by Strength)" button for non-draggers

v1.2 · Native slider, free wheel support (Round 3)

Question: "does it support wheel-to-add-weight like C4D's bar?" Conclusion: Blender's native slider already fine-tunes the value when hovered + wheel — no code needed. So the custom modal operator was deleted in favor of a native property settings.drag_bar (FloatProperty(subtype="FACTOR", min=-1, max=1, update=_on_drag_bar)):

- _on_drag_bar reads the current delta → dist = |val| * offset_delta → dispatch by mode to offset/smooth
- After applying, drag_bar resets to 0 (assigning 0 in the callback exits early — no infinite loop) → the bar visually zeroes after every gesture, infinite stacking
- Big win: tests upgraded from "replicating the modal algorithm" to "assigning the real property and firing the real callback" — far closer to the real code path

v1.3 · Weight Paint mode overhaul (Round 4)

Question: "can I see the wireframe and use fill select in weight mode, like C4D?" This round uncovered a fatal bug present since v1.0:

- ★ In Weight Paint mode, context.mode reads "PAINT_WEIGHT", not "WEIGHT_PAINT"! Three places (_require_mesh_edit, panel can_edit, group_new) had it wrong, so every operator silently failed in real Weight Paint mode and the panel stayed greyed out. Fixed everywhere + regression test added
- Select-mask toggles: panel exposes Blender native use_paint_mask / use_paint_mask_vertex so you can see the wireframe and box-select in weight mode
- Fill Select dual-mode: Edit mode uses from_edit_mesh; other modes read/write a bmesh copy — press Shift+Q right in weight paint mode
- ★ hasattr(bpy.ops.mesh, "fill_select") is always True (bpy.ops dynamically generates attributes for any string) → the panel wrongly assumed the addon was installed and called the wrong idname. Switched to hasattr(bpy.types, "ClassName") plus dual detection (classic mesh.fill_select / extension bl_ext.fill_select_between)

v1.4.2 · Wheel & mask fixes (Round 5)

Feedback: "wheel does nothing, vertex mask is all black, what are the mask shortcuts?"

- ★ Blender fine-tunes number fields with Ctrl+wheel — a bare wheel does nothing. The hint went into the panel
- drag_bar set step=10 (float actual increment = step/100 → 0.1 per notch) for a better wheel feel
- Mask shortcuts documented: Alt+Left select face, Shift+Alt add, B box select, A all

v1.4.3 · Step reverted (Round 5 fix)

- ★ Lesson: step affects BOTH wheel stepping AND drag feel — a bigger step makes dragging coarser. The user immediately reported "not as good as the previous version"; reverted to default step=1 (fine drag), wheel keeps Ctrl + small steps

v1.5.0 · List percentage bar + Smooth radius + Symmetry (Round 6)

Closed the C4D gap in the order "list percentage bar → Smooth radius → real-time symmetry":

- List percentage bar: each row of WM_UL_VertexGroups.draw_item shows a small bar with "the average weight of the currently selected vertices on this group" (weight_tools.weight_stats)
- Smooth radius: smooth_weights gained a radius param (BFS along edges for N layers, averaging the neighborhood; radius=1 keeps old behavior); panel SMOOTH mode exposes radius 1–5
- Symmetry toggle: exposes context.tool_settings.weight_paint.use_symmetry_x/y/z (※ not at tool_settings top level — a top-level use_symmetry_x doesn't exist) — with symmetry on, painting the left side mirrors to the right

v1.6.0 · Normalize (Round 7)

Request: "add normalize — and C4D lets you lock joints' weights, right?" (locking already existed; this round wired Normalize to respect it):

- normalize_weights(obj, indices, vg_locked): locked groups keep their weights, the rest are rescaled so each vertex sums to 1; skips vertices whose locked weights already reach 1 or that have no editable groups
- ★ Lesson: Python implicit multi-line string concatenation must be inside

parentheses! bl_description = "line one, " followed by "line two" on the next line without parens → whole addon failed to enable (IndentationError). Fixed by wrapping in parentheses

v1.7.0 · Influence highlight + Joint Filter (Round 8)

Request: "click a joint → viewport highlights its influence" + "Joint Filter":

- Influence highlight: SpaceView3D.draw_handler_add + POST_VIEW + 3D_UNIFORM_COLOR shader + POINTS (7px orange dots) drawn on vertices with weight > 1e-4. Cached data source (key=(id(mesh), vg_idx), 0.35s throttle to avoid a full-mesh scan every frame; Edit mode reads live bmesh read-only to avoid redraw loops)
- Joint Filter: _joint_filter_flags pure function (only show joints affecting the selection + name search), called from the UIList filter_items
- ★ Probe (Blender 5.0): Edit mode has no native "color by vertex group" overlay property → a custom GPU overlay is the only way; gpu.state.point_size_set still works in 5.0

v1.7.x · Copy/Paste weights (Round 9) + three real bugs

Request: "copy/paste joint weights" (mirror already existed, so that was just communication):

- weight.copy / weight.paste + a module-level _copy_buffer list
- ★ set_weights can't be used for paste (its second argument is a single scalar applied to all points) — per-point values must go through _write_all(obj, vg_idx, list(zip(indices[:n], _copy_buffer[:n]))). The first version passed the whole list to set_weights; the background test caught TypeError: BMDeformVert[key] = x: assigned value not a number
- Bug A (real crash): _influence_coords's Edit-mode branch called d.get() where d is a BMLayerItem (no .get) — an AttributeError spamming every frame in real Edit mode. Fixed to v[d].get(vg_idx, 0.0). ★ The 0.35s cache fooled the probe — you must reset the cache key to test the Edit branch
- Bug B: the list percentage bar originally shared settings.ul_weight_preview — Blender renders button values only after all draw_item calls finish, so the shared property was overwritten by the last row → every row showed the same value. VertexGroup can't hold dynamic attributes → switched to row.split(factor, align=True) self-drawn ratio bars (value computed and split on the spot)
- Bug C: filter_items sampled selection even in OBJECT mode (stale selection produced a stale filtered list) → only sample in EDIT_MESH/PAINT_WEIGHT

v1.8.0 · Vertex Weight Table + Weight HUD (Round 10, current)

Request: "vertex weight numeric table" (selected points show every bone's weight, editable) + weight HUD (confirmed via AskUserQuestion: "table first, and HUD — both this round"):

- Vertex Weight Table: two linked UIList widgets — WM_UL_WeightVerts (selected vertices) + WM_UL_WeightRows (every bone for the active vertex, descending, zeros at the bottom, each row's slider directly editable)
- Each row is an independent WeightTableRow(PropertyGroup) (Bug B's lesson: independent instances in a CollectionProperty avoid the shared-FloatProperty overwrite trap)
- Write callback _on_table_weight → weight_tools.set_weights → _finish_edit; respects locks (locked rows greyed/read-only)
- Refresh strategy: _table_populating guard against loops + signature caching (_table_sel_sig/_table_active_vert/_table_last_groups) rebuilds only on change; deleting a group forces a rebuild via the group-count signature to prevent group_index drift
- Weight HUD: POST_PIXEL draw handler, obj.ray_cast to find the vertex under the cursor + blf text showing bone name: weight; the whole handler is wrapped in try/except so no draw error can crash the viewport
- ★ Round-10 self-review caught 4 real bugs (all fixed):
  1. WM_UL_WeightRows.draw_item wrongly used data.vertex_groups — the rows template_list is bound to settings, so every row would AttributeError in the GUI. Changed to context.active_object
  2. The weight_hud toggle wasn't in the panel — added
  3. HUD ray two problems: obj.ray_cast works in object-local coordinates (a world ray misses on translated/rotated objects — must invert matrix_world); and Blender 5.0's ray_cast returns a 4-tuple (result, location, normal, index), not the old docs' 5-tuple — the ValueError was swallowed by try/except so the HUD never drew. Confirmed by probe, fixed

4. Stale table values: external edits to the same vertex left the table showing old numbers. Added _read_vert_all (read all groups for one vertex in one pass) + per-frame consistency check, rebuild only on mismatch

3. Comparison with C4D

Capability	C4D Weight Manager	This addon	Notes
Joint (bone) list	Joints list	Vertex groups list (same list widget + lock [lock])	click = current group
Slider-set weight	Auto Weight	Absolute / Add / Subtract / Smooth	ABSOLUTE jump-to-value; add/subtract accumulate by drag distance
Lock-joint normalize	yes	Normalize + locked groups preserved	locked read-only, rescale the rest
Copy / Paste	yes	copy/paste weights	one-to-one in selection order
Mirror / Invert	yes	Commands mirror / invert	X/Y/Z axis
Influence visualization	show influence on joint	influence highlight (orange dots)	same data source
Viewport HUD	weight HUD	weight HUD (live value at cursor)	
Joint Filter	yes	Joint Filter (affecting-selection + name search)	
Select by weight	Fill Selection	select by weight (=0/>0/<1/=1/≈range)	
Fill selection	Fill Selection (between loops)	integrated Fill Select addon	
Symmetric paint	symmetric	weight-brush symmetry toggle (native)	

Verdict: C4D Weight Manager's core interaction surface is fully replicated; the remaining advanced commands (Remap/Blur) are in §6.

4. Comparison with Maya

Maya has no single Weight Manager panel; its weight capabilities live in Paint Skin Weights and the Component Editor:

Maya capability	Here	Notes
Component Editor (per-point weight numbers, directly editable)	Vertex Weight Table (v1.8.0)	select points → top list switches point, bottom list edits each bone via slider, descending, zeros at bottom
Paint Skin Weights brush	Auto Weight four modes + native brush	this addon's "panel + slider" is more precise than brushing; native brush still available for smudging
Per-vertex exact numbers	vertex table + weight HUD	read numbers / show-at-cursor
Set / Add / Scale tools	Absolute / Add-Subtract / Smooth	clear correspondence
Normalize (Maya forces by default)	Normalize + locked groups	

Verdict: Maya's most-celebrated "Component Editor exact weight editing" lands here as the Vertex Weight Table, complementing the C4D-style panel.

5. Technical architecture

5.1 Code layout

weight_manager/ ├── __init__.py # entry: bl_info + operators + N-panel + settings + draw handlers ├── weight_tools.py # core algorithm layer (bmesh deform layer, headless-testable) └── README.md

- Algorithm layer weight_tools.py: pure logic, no UI. _read_all/_write_all batch read/write the bmesh deform layer (read once, write once — avoids per-vertex Python<->C calls; the performance core)
- UI layer __init__.py: operators + VIEW3D_PT_WeightManager panel + WeightManagerSettings + two GPU draw handlers

5.2 Key technical decisions

1. Always go through the bmesh deform layer: Blender 5.0 forbids VertexGroup.add() in Edit mode — all reads/writes go through bm.verts.layers.deform.verify() (Edit mode: live bm then update_edit_mesh; other modes: bmesh copy + to_mesh). Weight painting works in Edit mode on every version
2. Native widgets first: wheel fine-tuning and slider dragging are built-in Blender behavior — use native sliders instead of hand-rolled modals wherever possible
3. CollectionProperty with per-row instances: multi-row data (the vertex table) must give each row its own PropertyGroup — a shared FloatProperty is overwritten by the last row (Bug B lesson)
4. Draw handlers fully wrapped in try/except: no draw error may crash the viewport
5. Performance: influence data is cached + throttled (0.35s) to avoid a full-mesh scan every frame

5.3 Cross-version compatibility (4.2+ / 5.x)

- VertexGroup.lock → renamed lock_weight in 4.2; _vg_locked() uses getattr
- VertexGroup.add() forbidden in Edit mode → unified bmesh deform layer
- VertexGroup.weight(i) raises RuntimeError when the vertex isn't in the group → _weight() returns 0 safely
- 5.0 ray_cast 4-tuple vs old docs' 5-tuple → unpack as 4-tuple uniformly

5.4 Testing strategy (all headless background)

- Algorithm unit tests: test/test_algorithm.py, 24/24 pass
- End-to-end tests: test/test_e2e.py, 73/73 pass (real addon environment, incl. real PAINT_WEIGHT mode regression)
- Install verification: test/test_install.py, 8/8 pass (zip install → detect → enable → register)
- Cross-version: same script PASSes on 4.2.16 / 5.0.1 / 5.1.0
- Extension verification: tmp/verify_extension.py — after read_factory_settings clears state, addon_utils.enable("bl_ext.user_default.weight_manager") PASSes
- Limits: GPU overlay rendering, UIList rendering/dragging, real wheel feel, HUD ray picking — these GUI behaviors can't be tested headless. They rely on code self-review (which has caught 7 real bugs) + user GUI confirmation

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6. Future directions

Remaining gaps, by priority:

1. Weight import/export: export weights to JSON/CSV, or copy weights between meshes (Maya's copy skin weights / C4D's transfer)
2. Remap: C4D Weight Tool's Remap curve — remap weight values through a curve (compress/stretch a range around a threshold)
3. Blur: C4D's Blur command — set to neighborhood average by a fixed amount (Smooth already moves toward the average by drag distance; Blur is "set to average, repeatable")
4. Face/point-granularity fill selection: Fill Select currently fills "faces between two loops"; extend to "fill inside a loop" (the full C4D Fill Selection shape)
5. Auto axis detection for mirror: infer the symmetry axis from the object's bounds / selected-points centroid to reduce manual axis picking

6. Multi-object batch: apply the same weight operations to several meshes at once (common when clothing/accessories are modeled separately)
7. Edit-mode auto-normalize hint: when selected vertices' weight sum drifts from 1, show a one-click normalize hint in the panel
8. Publish to extensions.blender.org: the store/ package is ready (manifest + flat zip + icon/featured images) — submit for review to release publicly

7. Summary

Item	Value
Version	v1.8.0
Iterations	10 rounds
Algorithm unit tests	24/24
End-to-end	73/73
Install verification	8/8
Cross-version	Blender 4.2.16 / 5.0.1 / 5.1.0 all PASS
Extension	official-format enable all PASS
Real bugs caught by self-review	7 (PAINT_WEIGHT in v1.3, A/B/C in round 9, 1/2/3/4 in round 10)

One line: ten rounds of "feedback → rework → headless verify" turned C4D's "panel + slider" painting experience and Maya's "per-point numbers" precision into a complete Blender sidebar addon — from v1.0's feature baseline to v1.8.0's Vertex Weight Table + Weight HUD, the core interaction gaps are all closed.